Organic Compounds – Carbon, Functional Groups, Biomolecules, and Reactions

Organic chemistry is the chemistry of carbon compounds. It is the chemistry of life, of fuels, of plastics, of medicines, and of millions of substances that surround us every day. What makes carbon so special? Why can it form more compounds than all other elements combined? The answer lies in carbon's electronic structure and its unique ability to bond with itself and with other elements in endless combinations. From the simplest hydrocarbon — methane — to the most complex protein, all organic compounds share a common foundation: the carbon atom. Understanding carbon's bonding patterns, functional groups, biomolecules, and common reactions is essential for anyone who wants to understand the chemistry of living things and the materials of modern life.

This reading material explains why the carbon atom is the basis of organic chemistry by referring to its electronic structure and types of bonds formed. It explains the differences among the major groups of organic compounds — alkanes, alkenes, alkynes, haloalkanes, and alcohols — based on bonding patterns and functional groups. It provides the general formulas and elements present in carbohydrates, lipids, proteins, and nucleic acids, and explains why some are macromolecules and others are polymers. Finally, it describes simple reactions of organic compounds, including combustion, addition, condensation, and saponification of fats.

Part 1: Why Carbon Is the Basis of Organic Chemistry

The Electronic Structure of Carbon

Carbon has an atomic number of 6. Its electron configuration is **$1s^2\ 2s^2\ 2p^2$**. This means carbon has **four valence electrons** in its outermost energy level. To achieve a stable octet (eight valence electrons), carbon needs four more electrons. It achieves this by forming **four covalent bonds** — sharing electrons with other atoms.

Property	Value
Atomic number	6
Electron configuration	$1s^2\ 2s^2\ 2p^2$
Valence electrons	4
Bonds formed	4 covalent bonds
Electronegativity	2.55 (moderate)
Atomic radius	Small

Why Carbon Is Unique

Reason	Explanation
Tetravalency	Carbon forms four covalent bonds, allowing complex branching and diversity
Catenation	Carbon bonds with itself to form long chains, branched chains, and rings
Bond strength	C–C and C–H bonds are strong and stable
Multiple bonds	Carbon forms single, double, and triple bonds

Reason	Explanation
Moderate electronegativity	Carbon shares electrons fairly with many elements
Isomerism	The same molecular formula can produce different structures with different properties

Types of Bonds Carbon Forms

Bond Type	Example	Description
Single bond (C–C)	Ethane (CH_3–CH_3)	One pair of shared electrons; sp^3 hybridized
Double bond (C=C)	Ethene (CH_2=CH_2)	Two pairs of shared electrons; sp^2 hybridized
Triple bond (C≡C)	Ethyne (HC≡CH)	Three pairs of shared electrons; sp hybridized
Carbon–hydrogen (C–H)	Methane (CH_4)	Single bond with hydrogen
Carbon–oxygen (C–O)	Ethanol (CH_3CH_2OH)	Single bond with oxygen
Carbon–nitrogen (C–N)	Amines	Single bond with nitrogen
Carbon–halogen (C–X)	Chloromethane (CH_3Cl)	Single bond with halogen

Bond Hybridization and Geometry

Hybridization	Bonds Around Carbon	Geometry	Bond Angle	Example
sp^3	4 single bonds	Tetrahedral	109.5°	Methane (CH_4)
sp^2	1 double + 2 single	Trigonal planar	120°	Ethene ($CH_2=CH_2$)
sp	1 triple + 1 single	Linear	180°	Ethyne (HC≡CH)

Key insight: Carbon's four valence electrons, its ability to bond with itself, and its capacity to form single, double, and triple bonds make it the foundation of organic chemistry. No other element can form such a vast number of stable compounds.

Part 2: Major Groups of Organic Compounds

Hydrocarbons and Their Derivatives

Hydrocarbons are organic compounds made only of carbon and hydrogen.
Hydrocarbon derivatives are hydrocarbons in which one or more hydrogen atoms are replaced by other atoms or groups (such as halogens or the hydroxyl group).

Comparison of Major Groups

Group	General Formula	Functional Group	Bonding Pattern	Hybridization	Example
Alkane	C_nH_{2n+2}	None (only C–C and C–H)	All single bonds	sp^3	Methane (CH_4)
Alkene	C_nH_{2n}	C=C (double bond)	At least one double bond	sp^2	Ethene ($CH_2=CH_2$)
Alkyne	C_nH_{2n-2}	C≡C (triple bond)	At least one triple bond	sp	Ethyne ($HC\equiv CH$)
Haloalkane	R–X (X = F, Cl, Br, I)	–X (halogen)	Single bonds; halogen replaces H	sp^3	Chloromethane (CH_3Cl)
Alcohol	R–OH	–OH (hydroxyl)	Single bonds; OH replaces H	sp^3	Ethanol (CH_3CH_2OH)

Alkanes

Feature	Description
General formula	C_nH_{2n+2}
Bond type	All single bonds (saturated)

Feature	Description
Functional group	None
Reactivity	Relatively unreactive; undergo combustion and substitution
Examples	Methane (CH_4), ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10})
Uses	Fuels (LPG, gasoline, diesel), lubricants, waxes

Naming the first four alkanes:

Name	Formula	Carbon Atoms
Methane	CH_4	1
Ethane	C_2H_6	2
Propane	C_3H_8	3
Butane	C_4H_{10}	4

Alkenes

Feature	Description
General formula	C_nH_{2n} (for one double bond)

Feature	Description
Bond type	At least one C=C double bond (unsaturated)
Functional group	C=C
Reactivity	More reactive than alkanes; undergo addition reactions
Examples	Ethene (C_2H_4), propene (C_3H_6), butene (C_4H_8)
Uses	Production of plastics (polyethylene), ripening of fruits (ethene)

Alkynes

Feature	Description
General formula	C_nH_{2n-2} (for one triple bond)
Bond type	At least one C≡C triple bond (unsaturated)
Functional group	C≡C
Reactivity	Highly reactive; undergo addition reactions
Examples	Ethyne (C_2H_2), propyne (C_3H_4), butyne (C_4H_6)
Uses	Welding (oxyacetylene torch), production of plastics

Haloalkanes

Feature	Description
General formula	R–X (X = F, Cl, Br, I)
Functional group	–X (halogen)
Bonding	Halogen replaces a hydrogen atom
Reactivity	Undergo substitution and elimination reactions
Examples	Chloromethane (CH_3Cl), bromoethane (CH_3CH_2Br), chloroform ($CHCl_3$)
Uses	Solvents, refrigerants, anesthetics, pesticides

Alcohols

Feature	Description
General formula	R–OH
Functional group	–OH (hydroxyl)
Bonding	OH replaces a hydrogen atom
Reactivity	Undergo combustion, dehydration, oxidation, esterification

Feature	Description
Examples	Methanol (CH_3OH), ethanol (CH_3CH_2OH), glycerol ($C_3H_5(OH)_3$)
Uses	Solvents, fuels, disinfectants, beverages

Summary Comparison Table

Feature	Alkane	Alkene	Alkyne	Haloalkane	Alcohol
General formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}	R–X	R–OH
Functional group	None	C=C	C≡C	–X	–OH
Bond type	Single	Double	Triple	Single	Single
Saturation	Saturated	Unsaturated	Unsaturated	Saturated	Saturated
Hybridization	sp^3	sp^2	sp	sp^3	sp^3
Reactivity	Low	Moderate	High	Moderate	Moderate

Part 3: The Four Major Classes of Biomolecules

What Are Biomolecules?

Biomolecules are organic compounds that are essential to life. There are four major classes: carbohydrates, lipids, proteins, and nucleic acids. Each class has a characteristic composition, structure, and function.

Table of Biomolecules

Biomolecule	Elements Present	General Formula / Building Block	Monomer	Polymer	Function
Carbohydrates	C, H, O	$(CH_2O)_n$	Monosaccharide	Polysaccharide	Energy source; structural support
Lipids	C, H, O (sometimes P, N)	Variable (not a single formula)	Glycerol + fatty acids	Not a polymer	Energy storage; insulation; cell membranes
Proteins	C, H, O, N (sometimes S)	Variable (20 amino acids)	Amino acid	Polypeptide / protein	Structure; enzymes; transport; defense

Biomolecule	Elements Present	General Formula / Building Block	Monomer	Polymer	Function
Nucleic Acids	C, H, O, N, P	Variable (4 nucleotides)	Nucleotide	DNA, RNA	Storage and transmission of genetic information

Detailed Description of Each Biomolecule

Carbohydrates

Feature	Description
Elements	Carbon, hydrogen, oxygen
General formula	$(CH_2O)_n$ (ratio of H to O is 2:1)
Monomers	Monosaccharides (e.g., glucose, fructose, galactose)
Polymers	Disaccharides (e.g., sucrose, lactose), polysaccharides (e.g., starch, glycogen, cellulose)
Functions	Primary energy source; structural support (cellulose in plants)
Examples	Glucose, sucrose, starch, cellulose, glycogen

Lipids

Feature	Description
Elements	Carbon, hydrogen, oxygen (sometimes phosphorus and nitrogen)
General formula	Variable; not a single formula because lipids are diverse
Building blocks	Glycerol and fatty acids (in triglycerides)
Monomers	None (lipids are not polymers)
Functions	Long-term energy storage; insulation; protection; cell membrane structure; hormones
Examples	Fats, oils, phospholipids, steroids, waxes

Proteins

Feature	Description
Elements	Carbon, hydrogen, oxygen, nitrogen (sometimes sulfur)
General formula	Variable (20 different amino acids)
Monomers	Amino acids
Polymers	Polypeptides, proteins

Feature	Description
Functions	Structure (collagen, keratin); enzymes; transport (hemoglobin); defense (antibodies); signaling (hormones)
Examples	Hemoglobin, insulin, collagen, enzymes

Nucleic Acids

Feature	Description
Elements	Carbon, hydrogen, oxygen, nitrogen, phosphorus
General formula	Variable (4 different nucleotides)
Monomers	Nucleotides (sugar + phosphate + nitrogenous base)
Polymers	DNA (deoxyribonucleic acid), RNA (ribonucleic acid)
Functions	Storage and transmission of genetic information; protein synthesis
Examples	DNA, RNA, ATP

Macromolecules vs. Polymers

Term	Definition	Examples
Macromolecule	A large molecule made of many atoms; may or may not be made of repeating units	Carbohydrates, lipids, proteins, nucleic acids
Polymer	A large molecule made of repeating monomer units	Polysaccharides, proteins, nucleic acids

Key distinction:

Biomolecule	Macromolecule?	Polymer?	Reason
Carbohydrates	Yes	Yes (polysaccharides)	Made of repeating monosaccharide units
Lipids	Yes	No	Not made of repeating monomer units; built from glycerol and fatty acids
Proteins	Yes	Yes	Made of repeating amino acid units
Nucleic Acids	Yes	Yes	Made of repeating nucleotide units

Key insight: All polymers are macromolecules, but not all macromolecules are polymers. Lipids are macromolecules but not polymers because they do not consist of repeating monomer units.

Part 4: Common Reactions of Organic Compounds

1. Combustion

Combustion is the reaction of an organic compound with oxygen to produce carbon dioxide, water, and energy. It is an exothermic reaction.

Feature	Description
General equation	Organic compound + O_2 → CO_2 + H_2O + energy
Type	Exothermic (releases heat)
Example (methane)	CH_4 + 2 O_2 → CO_2 + 2 H_2O + energy
Example (ethanol)	C_2H_5OH + 3 O_2 → 2 CO_2 + 3 H_2O + energy
Importance	Fuels for cooking, transportation, and electricity

2. Addition Reactions

Addition reactions occur when atoms or groups are added to a molecule, typically breaking a double or triple bond. Alkenes and alkynes undergo addition reactions.

Type of Addition	Description	Example
Hydrogenation	Addition of H_2 to a double bond	CH_2=CH_2 + H_2 → CH_3–CH_3

Type of Addition	Description	Example
Halogenation	Addition of a halogen (e.g., Br_2)	$CH_2=CH_2 + Br_2 \rightarrow CH_2Br–CH_2Br$
Hydrohalogenation	Addition of HX (HBr, HCl)	$CH_2=CH_2 + HBr \rightarrow CH_3–CH_2Br$
Hydration	Addition of water (H_2O)	$CH_2=CH_2 + H_2O \rightarrow CH_3–CH_2OH$

Markovnikov's Rule: When HX is added to an unsymmetrical alkene, the hydrogen attaches to the carbon with more hydrogens, and the halogen attaches to the carbon with fewer hydrogens.

3. Condensation Reactions

Condensation reactions occur when two molecules combine with the loss of a small molecule, usually water. They are important in the formation of polymers and biomolecules.

Type of Condensation	Description	Example
Esterification	Alcohol + carboxylic acid → ester + water	$CH_3OH + CH_3COOH \rightarrow CH_3COOCH_3 + H_2O$
Peptide bond formation	Amino acid + amino acid → dipeptide + water	Glycine + alanine → dipeptide + H_2O
Glycosidic bond formation	Monosaccharide + monosaccharide → disaccharide + water	Glucose + fructose → sucrose + H_2O
Polymerization	Monomers join with loss of water	Formation of nylon, polyester

4. Saponification

Saponification is the hydrolysis of a fat or oil (triglyceride) with a strong base (such as NaOH or KOH) to produce soap (fatty acid salt) and glycerol.

Feature	Description
General equation	Fat + NaOH → Soap + Glycerol
Reactants	Triglyceride (fat or oil) and sodium hydroxide
Products	Soap (sodium salt of fatty acid) and glycerol
Type	Base-catalyzed hydrolysis
Importance	Traditional soap-making; production of biodiesel and glycerin

Example:

Triglyceride + 3 NaOH → 3 Soap (R–COONa) + Glycerol

Comparison of Common Organic Reactions

Reaction	What Happens	Key Feature	Example
Combustion	Organic compound burns in O_2	Releases CO_2, H_2O, and energy	Burning of methane
Addition	Atoms added across a double or triple bond	Unsaturated → saturated	Hydrogenation of ethene

Reaction	What Happens	Key Feature	Example
Condensation	Two molecules combine with loss of small molecule	Usually water is released	Esterification
Saponification	Fat + base → soap + glycerol	Base-catalyzed hydrolysis	Making soap from lard

Key Terms

Term	Definition
Organic chemistry	The study of carbon-containing compounds
Hydrocarbon	An organic compound made only of carbon and hydrogen
Alkane	A saturated hydrocarbon with only single bonds
Alkene	An unsaturated hydrocarbon with at least one double bond
Alkyne	An unsaturated hydrocarbon with at least one triple bond
Haloalkane	A hydrocarbon derivative in which a halogen replaces a hydrogen
Alcohol	A hydrocarbon derivative containing the hydroxyl (–OH) group
Functional group	A specific group of atoms that gives a compound its characteristic properties
Biomolecule	An organic compound essential to life
Carbohydrate	A biomolecule made of C, H, O with the general formula $(CH_2O)_n$
Lipid	A biomolecule made of C, H, O (sometimes P, N); not a polymer

Term	Definition
Protein	A biomolecule made of amino acid monomers
Nucleic acid	A biomolecule made of nucleotide monomers
Macromolecule	A large molecule made of many atoms
Polymer	A large molecule made of repeating monomer units
Combustion	Reaction with oxygen producing CO_2, H_2O, and energy
Addition reaction	Reaction in which atoms are added across a double or triple bond
Condensation reaction	Reaction in which two molecules combine with loss of a small molecule
Saponification	Hydrolysis of a fat with a base to produce soap and glycerol

Review Questions

1. Explain why carbon is the basis of organic chemistry. Refer to its electronic structure and types of bonds formed.

2. Differentiate alkanes, alkenes, and alkynes based on their general formula, bonding pattern, and functional group.

3. Complete the following table:

Group	General Formula	Functional Group	Example
Alkane			
Alkene			
Alkyne			
Haloalkane			
Alcohol			

4. Write the general formula for and list the elements present in each of the following biomolecules:

- (a) Carbohydrates
- (b) Lipids
- (c) Proteins
- (d) Nucleic acids

5. Explain why lipids are considered macromolecules but not polymers. Provide one other example of a biomolecule that is both a macromolecule and a polymer.

6. Describe each of the following reactions and provide one example:

- (a) Combustion
- (b) Addition
- (c) Condensation
- (d) Saponification

7. Write the balanced chemical equation for the complete combustion of ethane (C_2H_6).

8. Pag-aralan ang sitwasyon sa ibaba at sagutin ang mga tanong.

A student is comparing three unknown liquids. Liquid A boils at 78°C and is miscible with water. Liquid B boils at 56°C and evaporates quickly. Liquid C is a solid fat that reacts with NaOH to form a soapy substance.

- (a) What type of organic compound is Liquid A? Explain.
- (b) What type of organic compound is Liquid B? Explain.
- (c) What reaction occurs when Liquid C reacts with NaOH?

9. Using secondary sources, explain how the functional group of an alcohol differs from that of a haloalkane. Provide one example of each.

10. Application question: A company wants to produce soap from coconut oil. Describe the chemical reaction involved (saponification), the reactants and products, and why this reaction is important in industry.

Solutions and Answers

1. Carbon has four valence electrons ($1s^2\ 2s^2\ 2p^2$), allowing it to form four covalent bonds. It can bond with itself (catenation) to form chains, branches, and rings. It forms single, double, and triple bonds. These properties allow carbon to form millions of stable compounds, making it the basis of organic chemistry.

2. Alkanes have the general formula C_nH_{2n+2}, contain only single bonds (saturated), and have no functional group. Alkenes have the general formula C_nH_{2n}, contain at least one C=C double bond (unsaturated), and have the C=C functional group. Alkynes have the general formula C_nH_{2n-2}, contain at least one C≡C triple bond (unsaturated), and have the C≡C functional group.

3.

Group	General Formula	Functional Group	Example
Alkane	C_nH_{2n+2}	None	Methane (CH_4)
Alkene	C_nH_{2n}	C=C	Ethene ($CH_2=CH_2$)
Alkyne	C_nH_{2n-2}	C≡C	Ethyne (HC≡CH)
Haloalkane	R–X	–X	Chloromethane (CH_3Cl)
Alcohol	R–OH	–OH	Ethanol (CH_3CH_2OH)

4.
(a) Carbohydrates: $(CH_2O)_n$; elements C, H, O.
(b) Lipids: variable; elements C, H, O (sometimes P, N).
(c) Proteins: variable; elements C, H, O, N (sometimes S).
(d) Nucleic acids: variable; elements C, H, O, N, P.

5. Lipids are macromolecules because they are large molecules, but they are not polymers because they are not made of repeating monomer units. They are built from glycerol and fatty acids, which do not repeat in a chain. Proteins are both macromolecules and polymers because they are made of repeating amino acid units.

6.
(a) Combustion: reaction with oxygen producing CO_2, H_2O, and energy. Example: $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$.
(b) Addition: atoms added across a double or triple bond. Example: $CH_2=CH_2 + H_2 \rightarrow CH_3–CH_3$.
(c) Condensation: two molecules combine with loss of a small molecule (usually water). Example: $CH_3OH + CH_3COOH \rightarrow CH_3COOCH_3 + H_2O$.
(d) Saponification: hydrolysis of a fat with a base to produce soap and glycerol. Example: Fat + NaOH → Soap + Glycerol.

7. $2\ C_2H_6 + 7\ O_2 \rightarrow 4\ CO_2 + 6\ H_2O$ + energy

8.
(a) Liquid A is ethanol. It boils at 78°C, which matches ethanol's boiling point, and it is miscible with water because it has a hydroxyl group that forms hydrogen bonds.
(b) Liquid B is acetone. It boils at 56°C and evaporates quickly because it has weaker dipole-dipole forces and no hydrogen bonding with itself.
(c) Saponification occurs when Liquid C (a fat) reacts with NaOH to form soap and glycerol.

9. The functional group of an alcohol is the hydroxyl group (–OH), while the functional group of a haloalkane is a halogen atom (–X, where X = F, Cl, Br, or I). Example of an alcohol: ethanol (CH_3CH_2OH). Example of a haloalkane: chloromethane (CH_3Cl).

10. Saponification is the hydrolysis of a fat or oil with a strong base. The reactants are coconut oil (a triglyceride) and sodium hydroxide (NaOH). The products are soap (sodium salts of fatty acids) and glycerol. This reaction is important in industry because it is the basis of soap-making. It converts cheap fats and oils into valuable cleaning products and produces glycerol as a useful by-product.